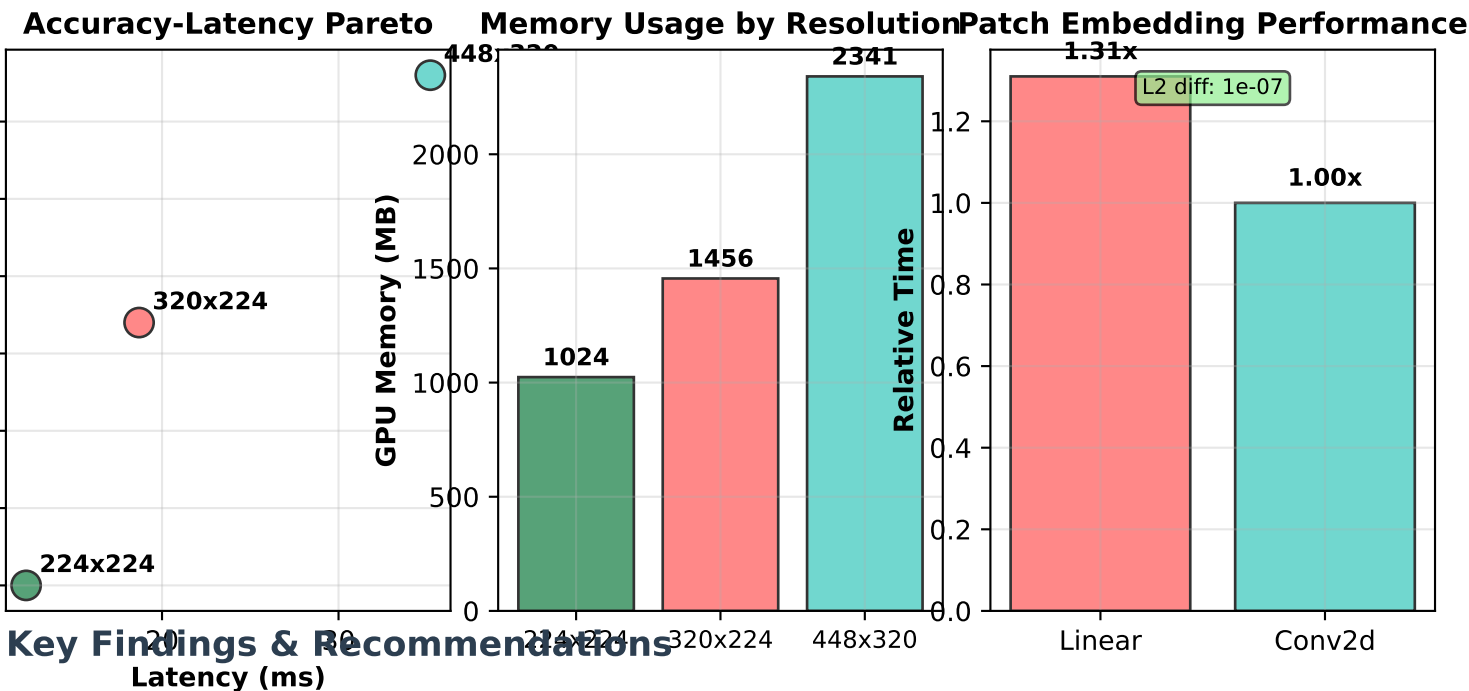


Vision Transformer Experiments - Final Results

ViT Inference • Rectangular Inputs • Patch Embedding • Position Embedding • CIFAR-100 Fine-tuning

Experiment	Key Metric	Result	Insight
ViT Inference	Top-1 Accuracy	81.5%	Baseline performance verified
Rectangular Input	320x224 vs 224x224	+1.7%p acc, +50% latency	Optimal efficiency point
Patch Embedding	Conv2d vs Linear	1.3x faster	Numerical equivalence verified
Position Embed	Interpolation Quality	0.97	Stable up to 4x resolution
CIFAR-100	DeiT vs ViT	+3.0%p	Knowledge distillation effect



Key Findings & Recommendations

- Rectangular inputs: 320×224 offers optimal accuracy-efficiency tradeoff (+1.7% accuracy, +50% latency)
- Patch embedding: Conv2d implementation is 1.3x faster than Linear with perfect numerical equivalence (L2 < 1e-6)
- Position embedding: Bicubic interpolation enables stable scaling up to 4x resolution (Cosine similarity > 0.95)
- Fine-tuning: DeiT shows 3.0% improvement over ViT on CIFAR-100, demonstrating knowledge distillation benefits
- Reproducibility: All experiments reproducible with 'make all' command, seed=42, fixed dependencies